Econometrics

Binary Choice and Selection Models

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Outline

1. Maximum Likelihood
2. Models of Discrete Choice
3. Sample Selection Models

Presentation

- An ordinary linear model can be estimated via OLS
- But there exists another method: *Maximum Likelihood*
- We write the *likelihood* of the sample, which we maximize with respect to the parameters we'd like to estimate
- We then get the optimal parameters
- In the OLS method, we minimized the sum of squared residuals: it is again an optimization process

The model (simple case)

- Let's consider an ordinary linear model $y_i = a + bx_i + u_i$
- Assume that the error term u_i follows a normal distribution $N(0, \sigma^2)$
- We consider a, b, and x as fixed
- Thus y_i follows a normal distribution as well: $N(a + bx_i, \sigma^2)$
- The "likelihood" of the sample is the joint probability (density) of the observed sample, seen as a function of the parameters

Estimation

- The distribution of y depends on parameters a and b that are unknown
- If we think that the sample was correctly sampled, then parameters a and b are such that the probability of having sampled this particular data is maximum
- In other words, these parameters should maximize the likelihood of the sample
- This is the ML method: Maximum Likelihood

Computation

- We assume that individuals in the sample are independent
- The likelihood of the sample is thus the product of the likelihood of each individual
- For this reason, we usually take its log to get a sum (easier to handle)
- We thus deal with the "log-likelihood"
- Parameters a and b are obtained by maximizing the log-likelihood, which is the same as maximizing the likelihood since the log function is strictly increasing
- In practice: usual maximization techniques, or iterative methods
- With ML, we get the same estimates of a and b as with OLS, with $\hat{\sigma}^2$ biased downwards ($\hat{\sigma}^2_{ML} = SSR/N$)

Application

- $\forall i,\ y_i \hookrightarrow N(a + bx_i, \sigma^2)$, with density $f(y_i) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}(y_i - a - bx_i)^2\right)$
- So the likelihood of the sample is $L = \prod f(y_i)$, and the log-likelihood is:

$$LL = \log(L) = \sum \log(f(y_i)) = -\frac{N}{2}\log(2\pi) - N\log\sigma - \frac{1}{2\sigma^2}\sum(y_i - a - bx_i)^2$$

- In the end: $\max_{(a,b)} LL \iff \min_{(a,b)} \sum(y_i - a - bx_i)^2$: in this special case, OLS and ML estimators of a and b are identical
- Similarly, we find $\hat{\sigma}^2_{ML} = \frac{SSR}{N}$, which is biased downwards (we found earlier that an unbiased estimator was $\hat{\sigma}^2_{OLS} = \frac{SSR}{N-k}$)
- However, although biased, it is consistent, since in a large sample N and $N - k$ are almost the same

Maximum Likelihood: General Properties

Under regularity conditions:

- ML estimators can be biased
- But they are consistent
- Asymptotically normally distributed
- Asymptotically efficient (they reach the Cramér-Rao bound: smallest asymptotic variance among consistent asymptotically normal estimators)
- Caution: the likelihood must be well specified

Tests

No OLS tests any more, since we are no longer in the OLS framework. Assume we want to test q linear restrictions on parameter θ: $R\theta = r$. The possibilities, which are asymptotically equivalent, are:

- Wald test: estimate parameter θ by ML, then check whether $R\hat{\theta} - r$ is close to zero, using its asymptotic variance-covariance matrix (this procedure is close to the tests we are used to)
- Likelihood ratio test: estimate the model with and without the constraints, and check whether the difference between the 2 LL is significantly different from 0:
  $LR = 2(LL_{unconstrained} - LL_{constrained}) \hookrightarrow \chi^2_q$ under H_0
- Lagrange multiplier test: estimate the model with the restrictions, then check whether the first order condition of the unrestricted model is significantly violated

Binary Choice

- Example: for individual i, $y_i = 1$ if unemployed, $y_i = 0$ if employed
- We want to explain y by age, education, etc.
- OLS fits a line, but here there is no real scatterplot since y can take only 2 values, 0 and 1
- Plus, many other problems, the main one being that predictions will be out of range
- So the preferred method is ML

Discrete Choice Models and OLS (1)

- Example: for individual i, $y_i = 1$ if unemployed, $y_i = 0$ if employed
- We want to explain y by age and education (the *linear probability model*):
- $y_i = \alpha + \beta\,\text{age}_i + \gamma\,\text{education}_i + \varepsilon_i = x_i b + \varepsilon_i$
- So $E(y_i|x_i) = x_i b$
- And we know that $E(y_i|x_i) = 1 \times P(y_i = 1|x_i) + 0 \times P(y_i = 0|x_i)$, so $P(y_i = 1|x_i) = x_i b$
- $x_i b$ should lie between 0 and 1, which imposes strong restrictions on x and $b \Rightarrow$ inconvenient

Discrete Choice Models and OLS (2)

- Furthermore, ε_i can take only 2 values: $-x_i b$ with probability $1 - x_i b$, or $1 - x_i b$ with probability $x_i b$
- So that: $V(\varepsilon_i) = x_i b(1 - x_i b)$
- There is heteroskedasticity

Discrete Choice Models and OLS (3)

- Heteroskedasticity could be dealt with using White's standard errors (see lecture on FGLS)
- Non-normality could be dealt with using a large sample size (Central Limit Theorem)
- But the main problem is that we need $E(y_i|x_i)$ to belong to $[0, 1]$
- For these various reasons, OLS is inappropriate for predicting probabilities (although the linear probability model with robust standard errors is still often used to estimate average effects)

Solution

- We act as if there was an unobserved continuous variable y^* such that $\forall i$, $y_i^* > 0 \Rightarrow y_i = 1$ and $y_i^* \leq 0 \Rightarrow y_i = 0$
- This new variable y^* is continuous: we can thus refer to our usual model, with some adaptations

The Model

- Assume that $y^* = X\beta + u$
- Let F be the cumulative distribution function (cdf) of u, with a distribution symmetric around 0
- $P(y_i = 1) = P(y_i^* > 0) = P(X_i\beta + u_i > 0) = P(u_i > -X_i\beta) = P(u_i < X_i\beta) = F(X_i\beta)$
- Usual functions for F are the normal (*Probit model*) or the logistic (*Logit model*)

Distributions

- Normal: $F(x) = \Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$
- Logistic: $F(x) = \frac{1}{1+e^{-x}} = \frac{e^x}{1+e^x}$
- These 2 are very close, and give similar results
- Expectation: 0 for both distributions
- Variance: 1 for the standard normal, $\pi^2/3$ for the logistic
- Characteristics of the *Logit*: extreme events have higher probability (heavier tails); parameters are interpreted more easily (odds-ratios, see below)

Log-likelihood (1)

Likelihood function for the entire sample:

$$L = \prod P[y_i = 1|x_i, \beta]^{y_i} P[y_i = 0|x_i, \beta]^{1-y_i}$$

So that when $y_i = 0$, only the right-hand-side part is there, and when $y_i = 1$, only the left-hand-side part is there.
And we have $P[y_i = 1|x_i, \beta] = P[y_i^* > 0|x_i, \beta] = F(x_i\beta)$, so

$$LL = \sum y_i \log(F(x_i\beta)) + \sum (1 - y_i) \log(1 - F(x_i\beta))$$

Maximizing this log-likelihood amounts to differentiating this expression with respect to parameter β and setting it to zero (it can be shown that it is globally concave for the Probit and the Logit).

Log-likelihood (2)

$$LL = \sum y_i \log(F(x_i\beta)) + \sum (1 - y_i) \log(1 - F(x_i\beta))$$

The first order condition is the following (derivative wrt β is 0): $\frac{\partial LL}{\partial \beta} = 0$, which in fact is a column vector of derivatives (derivative of LL wrt β_1, β_2, etc.)

$$\frac{\partial LL}{\partial \beta} = \sum \left[\frac{y_i - F(x_i\beta)}{F(x_i\beta)(1 - F(x_i\beta))} f(x_i\beta) \right] x_i' = 0$$

For the logistic distribution, $F(x) = \frac{e^x}{1+e^x}$ and $f(x) = \frac{e^x}{(1+e^x)^2} = F(x)(1 - F(x))$, so:

$$\frac{\partial LL}{\partial \beta} = \sum \left(y_i - \frac{\exp(x_i\beta)}{1 + \exp(x_i\beta)} \right) x_i' = 0$$

The optimal value for β is found with optimization techniques (iterative).

Remark: Scaling

- Recall that $P(y_i = 1) = P(u_i < X_i\beta) = F(X_i\beta)$: this holds only if u has a standardized distribution (of variance 1 for the Probit)
- u could in fact have *any* standard deviation σ; in that case:
  $P(u_i < X_i\beta) = P(u_i/\sigma < X_i\beta/\sigma) = F(X_i\beta/\sigma)$
- So the estimate we compute is really β/σ: there is no way to identify the two separately
- The convention is to set $\sigma = 1$ for the Probit model, and $\sigma = \pi/\sqrt{3}$ (the standard deviation of the logistic distribution) for the Logit model
- This multiplicative scaling changes neither the signs nor the ratios β_j/β_k of parameters, nor the predictions
- The only consequence is that Probit and Logit parameters are scaled differently:
  $\hat{b}_{logit} \simeq 1.6\,\hat{b}_{probit}$ in practice ($\pi/\sqrt{3} \simeq 1.8$ if we only match the variances)

Remark: Predicted and Actual Frequency

- $\hat{p}_i = \hat{P}(y_i = 1) = F(x_i\hat{\beta})$
- For the Logit, $\hat{p}_i = 1/(1 + \exp(-x_i\hat{\beta})) = \exp(x_i\hat{\beta})/(1 + \exp(x_i\hat{\beta}))$
- $\hat{\beta}$ is the solution of the first order condition, so that we get:

$$\sum \left(y_i - \frac{\exp(x_i\hat{\beta})}{1 + \exp(x_i\hat{\beta})} \right) x_i' = 0$$

$$\sum (y_i - \hat{p}_i)x_i' = 0 \quad \Longleftrightarrow \quad \sum \hat{p}_i x_i' = \sum y_i x_i'$$

If there is a constant term in x, then $\sum \hat{p}_i = \sum y_i$: the predicted frequency is exactly equal to the actual frequency. This holds approximately for the Probit model.

Interpreting Parameters (1)

Let x_i be a row vector of observations (corresponding to individual i) and β be a column vector of parameters.

- $P(y_i = 1) = F(x_i\beta)$
- Parameters enter non-linearly in the expression
- If we differentiate $F(x_i\beta)$ with respect to variable x_k, calling f the derivative of F, we get:

$$\frac{\partial F(x_i\beta)}{\partial x_{i,k}} = f(x_i\beta)\beta_k$$

With f either the standard normal density ϕ (*probit*) or the logistic density $\exp(x)/(1+\exp(x))^2$ (*logit*).
The effect of a change in $x_{i,k}$ depends on the values of x_i, so it is different for each person.

Interpreting Parameters (2)

$$\frac{\partial F(x_i\beta)}{\partial x_{i,k}} = f(x_i\beta)\beta_k$$

- The *sign* of the effect of a change in $x_{i,k}$ can however be determined
- Since $f > 0$, it is the sign of β_k

Marginal Effects

- We still miss a way to measure the effect of X on y: the *marginal effects*
- Consider the average person in the sample and compute the prediction $\hat{y}_0 = F(\bar{X}\hat{\beta})$
- Then increase variable x_1 by one unit, and compute the prediction again: $\hat{y}_1$
- The marginal effect of x_1 (at the mean) is $\hat{y}_1 - \hat{y}_0$: by how much the probability of $[y = 1]$ increases when x_1 increases by one unit (e.g. one more year of education, switching from male to female if gender is a dummy)
- Always mention that this was computed for the average person; it can also be computed for specific individuals (e.g. the average 40-year-old white-collar worker)
- Alternative: the *average marginal effect* (AME), i.e. the individual effects averaged over the whole sample
- statsmodels: `res.get_margeff(at="mean")`, or `at="overall"` for the AME

Logit vs. Probit

- Since the two distributions are quite similar, predictions will be similar as well
- However, values of parameters differ: this is because of the different scaling
- We usually get $\hat{b}_{logit} \simeq 1.6\, \hat{b}_{probit}$
- Matching only the variances would give: $\hat{b}_{logit} \simeq (\pi/\sqrt{3})\hat{b}_{probit} \simeq 1.8\, \hat{b}_{probit}$
- Again, this does not mean that the impact of explanatory variables is higher with the Logit model

Logit: Odds-ratios (1)

- Parameters β do not have a linear impact: we can compute marginal effects, but the Logit provides more information
- Recall that $P(y_i = 1) = P(y_i^* > 0) = F(X_i\beta) = \frac{1}{1+e^{-X_i\beta}}$
- I.e. $\forall i$, $P(y_i = 1) = p_i$ and $P(y_i = 0) = 1 - p_i$
- Let's define the *odds* c_i as $c_i = \frac{p_i}{1-p_i}$
- $[y_i = 1]$ is c_i times more likely to happen than $[y_i = 0]$
- We have $p_i = \frac{1}{1+e^{-X_i\beta}} = \frac{e^{X_i\beta}}{1+e^{X_i\beta}}$
- And $1 - p_i = \frac{1}{1+e^{X_i\beta}}$
- So $c_i = e^{X_i\beta}$

Logit: Odds-ratios (2)

- We thus get $c_i = e^{X_i \beta}$
- But $X_i \beta = \beta_0 + \sum_j x_{i,j} \beta_j$
- So $c_i = e^{\beta_0 + \sum_j x_{i,j} \beta_j} = e^{\beta_0} \prod_j e^{x_{i,j} \beta_j}$
- If variable $x_{i,k}$ increases by 1 unit, c_i is multiplied by e^{β_k}
- If variable x_k increases by 1 unit, it multiplies the *odds* that $[y_i = 1]$ happens by e^{β_k}
- We call *odds-ratio* associated with variable x_k the value e^{β_k} (in Python: `np.exp(res.params)`)

Goodness of Fit (1)

- Pseudo-R^2 (McFadden): $R^2 = \frac{LL_{fit} - LL_0}{LL_{max} - LL_0} = 1 - \frac{LL_{fit}}{LL_0}$ (since $LL_{max} = 0$)
- LL_{fit}: log-likelihood of the model
- LL_0: log-likelihood of a model with only a constant, where the probability is estimated by the proportion of ones in the sample
- LL_{max}: maximum log-likelihood attainable. In our case, 0, because a model that predicts the observed values perfectly has likelihood 1, and $\log(1) = 0$
- statsmodels reports it as "Pseudo R-squ."

Goodness of Fit (2)

- We can also evaluate the goodness of fit of the model by comparing correct and incorrect predictions
- Predicted outcomes: since $0 \leq \hat{p}_i \leq 1$, we need to choose a cut-off point c that will determine if the prediction will be 0 or 1 (usually: $c = 1/2$)
- *Sensitivity* (sensibilité): true positive rate in predictions
- *Specificity* (spécificité): true negative rate in predictions
- ROC (Receiver Operating Characteristic) curve: plots the fraction of $y = 1$ values correctly classified (true positives) against the fraction of $y = 0$ values incorrectly classified (false positives) as c varies, i.e. *sensitivity* against *(1 − specificity)*

The ROC Curve

- Comes from signal detection theory, developed during WW2
- The accuracy of the classification rule is measured with the area under the ROC curve (AUC)
- The area represents the ability of the procedure to correctly classify individuals, and lies between 0.5 (random classification) and 1 (perfect classification)
- A random classification rule yields a diagonal ROC curve
- Trade-off between sensitivity and specificity: the more bowed the curve is, the better the model is
- The closer the ROC curve to the upper-left corner, the higher the predictive power
- The optimal cut-off point is determined by the analyst depending on the issue studied

Could Heteroskedasticity be an Issue?

- In Probit and Logit models, homoskedasticity of u is assumed in the latent variable framework (normalization: $\sigma = 1$)
- Recall that in the Probit model, only β/σ can be identified
- Heteroskedasticity means that σ is a function of some variables, say $\sigma_i = \exp(Z_i\gamma)$
- Remark 1: it is best not to have a constant in variables Z, because then testing $\gamma = 0$ amounts to testing $\sigma = \exp(0) = 1$, which means homoskedasticity
- Remark 2: variables Z should be exogenous variables not already in the model (see Cameron & Trivedi)
- In this case, we model $P(y_i = 1) = P(y_i^* > 0) = F(X_i\beta/\sigma_i)$ and plug in $\sigma_i = \exp(Z_i\gamma)$ in the likelihood
- Stata's *hetprobit* provides this estimation, with a test of $\gamma = 0$; in Python, the likelihood can be written with statsmodels' `GenericLikelihoodModel`

Could Endogeneity be an Issue?

- Just like OLS, Probit and Logit estimates are inconsistent if one or several explanatory variables are endogenous
- The usual latent variable model is written: $y^* = X\beta + x\gamma + u$, with x endogenous
- In that case, given we find enough instruments Z, the endogenous variable should be instrumented: $x = Zb + Xc + v$
- To estimate this now 2-equation model in the maximum likelihood framework, we only need to postulate that (u, v) follow a bivariate normal distribution and plug this into the likelihood
- Testing endogeneity of x amounts to testing whether u and v are uncorrelated
- All this requires both normality and homoskedasticity of u and v, so Probit is required (no Logit here)

About Selection

- E.g. consider the interpretation of average scores over time of an achievement test
- A decline over time may be due to real deterioration in student knowledge, or it may just reflect a selection effect, i.e. more students have been taking the test over time and the new test takers are the relatively weaker students
- Selection can arise from self-selection (individuals choose to participate or not in a particular activity) or sample selection (the sample is not random: e.g. those who participate in the activity are oversampled)
- These problems are treated alike in Econometrics through *sample selection models*

The Heckit Model (or Generalized Tobit)

- Say a variable y_2^* is partially observed (y_2^* is the latent variable)
- Say we have another latent variable y_1^*, such that y_2^* is observed only if $y_1^* > 0$
- E.g. health expenditures or wages
- y_1 is participation in the labor market: $y_1 = 0$ (does not participate) or $y_1 = 1$ (participates)
- y_2 is the observed wage, which cannot be observed if the person does not participate in the labor market: that person may have a positive potential wage y_2^*, but we only observe $y_1 = 0$
- We'll see here why the Probit model is useful for the estimation of the 1st step equation
- Note: in the *standard* Tobit model, y_2 is observed only if it is above (or below) a certain threshold

The Model

Latent variables: $y_1^* = x_1 b_1 + u_1$ and $y_2^* = x_2 b_2 + u_2$

- Participation equation: $y_1 = 1$ if $y_1^* > 0$, $y_1 = 0$ otherwise
- Outcome equation: $y_2 = y_2^*$ if $y_1 = 1$, y_2 is missing otherwise

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \hookrightarrow N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix} ; \begin{pmatrix} 1 & \sigma_{1,2} \\ \sigma_{1,2} & \sigma_2^2 \end{pmatrix} \right)$$

The Logit model could not be used here for the 1st equation, since the error terms are jointly normal: we have to consider the Probit model.

The Likelihood

$$L = \prod [P(y_{1,i}^* \leq 0)]^{1-y_{1,i}} [f(y_{2,i}|y_{1,i}^* > 0) \times P(y_{1,i}^* > 0)]^{y_{1,i}}$$

- First term: discrete contribution when $y_{1,i}^* \leq 0$
- Second term: continuous contribution when $y_{1,i}^* > 0$

Conditional Means

$$E(y_2|x, y_1^* > 0) = E(x_2 b_2 + u_2|x_1 b_1 + u_1 > 0) = x_2 b_2 + E(u_2|u_1 > -x_1 b_1)$$

In the normal case, it can be shown that:

$$E(y_2|x, y_1^* > 0) = x_2 b_2 + \sigma_{1,2}\lambda(x_1 b_1)$$

With λ the *inverse Mills ratio*: $\lambda(z) = \frac{\phi(z)}{\Phi(z)}$

- If u_1 and u_2 are independent, then $E(u_2|u_1 > -x_1 b_1) = 0$ and OLS are consistent in the second equation; otherwise, they are not: we have to take into account the sample selection bias $\sigma_{1,2}\lambda(x_1 b_1)$
- Error terms represent unobservable heterogeneity: it is likely that individual heterogeneity has an impact both on the decision to participate and on the outcome (e.g. health care expenditures)
- The sign of corr(u_1, u_2) is the sign of $\sigma_{1,2}$

The Heckman Two-Step Estimator

$$E(y_2|x, y_1^* > 0) = x_2 b_2 + \sigma_{1,2}\lambda(x_1 b_1)$$

Idea: first estimate λ, then estimate the model replacing λ by its estimate $\hat{\lambda}$

1. Probit regression of y_1 on x_1 (indeed, $P(y_1^* > 0) = \Phi(x_1 b_1)$)
2. Compute $\hat{\lambda} = \frac{\phi(x_1 \hat{b}_1)}{\Phi(x_1 \hat{b}_1)}$
3. OLS regression of y_2 on x_2 and $\hat{\lambda}$, on the selected sample only ($y_1 = 1$)
4. The coefficient of $\hat{\lambda}$ estimates $\sigma_{1,2}$: its sign gives the sign of the correlation between the 2 error terms, and if it is not significant, the 2 equations can be considered independent

Remarks on the two-step estimator

- Two-step Heckman estimates are consistent
- This method is easy and fast to implement
- But the usual OLS standard errors of the second step are wrong ($\hat{\lambda}$ is itself estimated): they must be corrected
- There is also an efficiency loss (not appropriate in small samples)
- A more widely used method is Maximum Likelihood on the full model
- Notice that the use of a Logit would not be appropriate here

Identification Considerations

- Theoretically, the exact same regressors can appear in both equations, because $\lambda(\cdot)$ is non-linear: no strict multicollinearity
- However, in that case, the model is close to unidentified, because the inverse Mills ratio is almost linear over a wide range of values
- This leads to a great instability of parameters (see slides on near-perfect multicollinearity)
- Thus, at least one exclusion restriction is usually required: at least one variable in the first equation (selection) should be absent from the second equation (conditional outcome)

Selection on observables vs. unobservables

- If we find out that there is in fact no correlation between the 2 equations (check $\rho = \text{corr}(u_1, u_2) = \sigma_{1,2}/\sigma_2$), they can be estimated independently through a Two-Part model (see next slide)
- If there is no correlation: selection is said to be made on observables, because it is fully explained by regressors
- If there is correlation: selection is said to be made on unobservables, because it is partly explained by error terms, which comprise all the unobservable information
- The Two-Part model can also be used if the correlation of the inverse Mills ratio with regressors is too strong (near-perfect multicollinearity)

The Two-Part Model

This model intentionally does not take into account potential correlation between the error terms of the 2 equations

1. Participation equation: Probit or Logit that gives the probability to participate
2. Conditional outcome equation: anything we like (linear model, count data, etc.) that gives the level of activity, given that the individual participates (warning: count models then need a distribution truncated at 0, such as the zero-truncated Poisson or negative binomial); so it is more flexible than the Generalized Tobit framework

And the prediction of the model is the product of the two: $E(y) = P(y > 0) \times E(y|y > 0)$. Notice that we lose the assessment of the selection effect but gain in flexibility (no need to stick to Normal models).